

STUDENT PERFORMANCE PREDICTION

Week 2 – EDA & Visualization Framework

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Revised to address evaluation feedback: concrete examples, numerical illustrations, diagrams embedded in the report, and more explicit feature-engineering and model-evaluation readiness.

Planning document • Illustrative values are assumptions, not experimental results.

1. Introduction to Exploratory Data Analysis (EDA)

Exploratory Data Analysis is the systematic process of inspecting, cleaning, summarizing, and visualizing a dataset before statistical modeling or machine learning. For the Student Performance Prediction project, EDA will identify data-quality issues and relationships among academic, attendance, behavioral, and demographic variables.

Project-scale planning example: the eventual dataset is planned as approximately **1,000 student records**. This is illustrative. If 40 attendance values were missing, the missing-value rate would be $40/1,000 \times 100 = 4\%$. The actual rate will be calculated after the dataset is obtained.

EDA can reveal missing values, duplicates, invalid ranges, skewed distributions, outliers, class imbalance, correlations, and possible leakage before modeling.

2. EDA Objectives

- Understand dimensions, variable types, ranges, distributions, and target balance.
- Measure missing values and duplicates using counts and percentages.
- Identify invalid values, inconsistent categories, extreme observations, and suspicious records.
- Study individual variables through univariate analysis.
- Compare predictors with the target through bivariate analysis.
- Investigate interactions and correlations through multivariate analysis.
- Generate evidence that guides preprocessing and later feature engineering.
- Document findings so each cleaning or transformation decision can be justified.

3. Expected Data Types

Data type	Possible project examples	EDA checks
Numerical	attendance, study hours, marks, absences	range, mean, median, spread, outliers
Categorical	gender, school support, activity participation	frequency, rare categories
Ordinal	performance level, parental education	ordering and category counts
Binary	internet access, extra support yes/no	class proportions
Date/Time	assessment date, enrollment date	validity, seasonality, gaps
Text	comments or feedback (if available)	length, missingness, duplicates

4. Initial Dataset Inspection

The first inspection stage establishes a reproducible baseline before any values are changed.

Example Python checks:

```
df.head() df.shape df.info() df.describe(include="all") df.dtypes df.isnull().sum() df.duplicated().sum()
```

Concrete interpretation example: if `df.shape` returns **(1000, 12)**, the dataset contains 1,000 rows and 12 columns. If attendance has 40 null values, its missingness is 4%. These figures are examples only.

5. Data Quality and Cleaning Framework

5.1 Missing Data

Missingness will be quantified per column before choosing an action. The strategy will depend on percentage missing, variable type, and the likely reason for missingness.

Situation	Illustrative example	Planned action
0–5% missing	attendance = 4% missing	median/mode imputation after checking distribution
5–30% missing	feature = 15% missing	investigate cause; compare imputation vs removal
>30% missing	feature = 45% missing	assess usefulness; consider removal if justified
Missing target	performance label unavailable	exclude from supervised training rows

The thresholds are planning guidelines, not universal rules; final treatment will be based on the actual dataset.

5.2 Duplicate and Invalid Records

Duplicates will be counted with `df.duplicated().sum()`. Domain validation will also be applied. For example, attendance and marks should normally be between **0 and 100**, while study hours should not be negative. Violations will be investigated rather than silently deleted.

5.3 Outlier Detection

Outliers will be examined using box plots and the IQR method: $IQR = Q3 - Q1$; lower bound = $Q1 - 1.5 \times IQR$; upper bound = $Q3 + 1.5 \times IQR$.

Illustrative example: if $Q1 = 55$ and $Q3 = 85$, $IQR = 30$ and the upper statistical boundary is $85 + 45 = 130$. A mark of 135 would be flagged statistically, but because marks should be 0–100, it also requires a domain-validity check.

6. Univariate Analysis

Univariate analysis studies one variable at a time and establishes its distribution and quality.

Variable	Recommended plot	Questions
Study hours	histogram + KDE	skewed? concentrated around a few values?
Marks	histogram / box plot	spread, median, extreme values?
Attendance	histogram / box plot	low-attendance group? outliers?
Gender / support	count plot	category imbalance?
Target class	bar chart	class balance?

Concrete example: if 1,000 planning records contain 700 'Not At Risk' and 300 'At Risk', the class proportions are 70% and 30%. This 70:30 split would be considered when choosing evaluation metrics. These are illustrative values.

7. Bivariate Analysis

Bivariate analysis studies relationships between two variables, with special attention to predictors versus the target.

Pair	Plot	Example question
Study hours × marks	scatter plot	does greater study time align with marks?
Attendance × marks	scatter plot	is low attendance associated with lower marks?
Support × target	grouped bar chart	does support status differ by risk class?
Activity × marks	box plot	do activity groups show different score distributions?
Gender × target	count/stacked bar	are class proportions different across groups?

Numerical example: if average assessment score is 78 and an earlier assessment score is 65, the gap is $78 - 65 = 13$ points. This can be explored as a possible derived feature only if the information is available before prediction and does not leak the outcome.

8. Multivariate Analysis

Multivariate analysis examines several variables simultaneously.

- Correlation heatmap for numerical features.
- Pair plots for a small set of high-value numerical variables.
- Colored scatter plots to inspect separation by target class.
- Grouped summaries such as support × attendance × target.
- Interaction candidates such as attendance × study hours and subject average × attendance.

Correlation does not establish causation; patterns observed in EDA must be validated during modeling.

9. Embedded EDA Workflow Diagram

Week 2 EDA Workflow for Student Performance Prediction

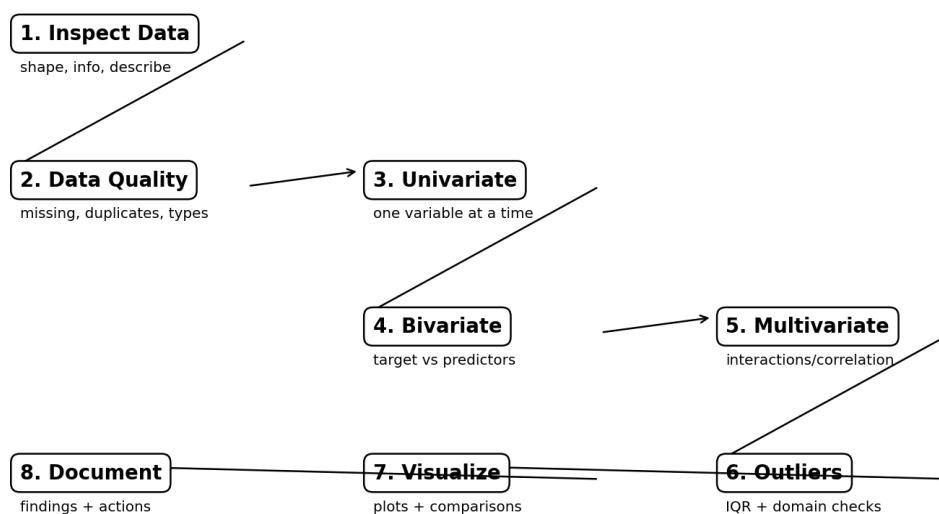


Figure 1. Planned EDA workflow from inspection through documentation.

10. Visualization Strategy

Visualization Strategy Mapped to EDA Questions

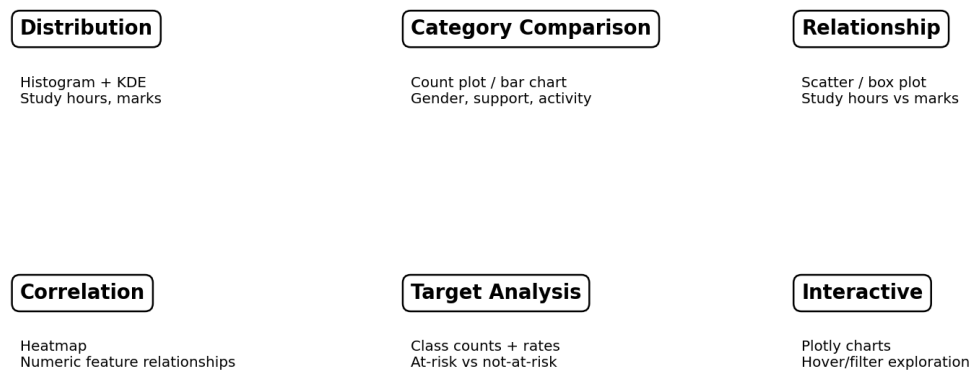


Figure 2. Visualization strategy mapped to common EDA questions.

EDA question	Primary visualization	Secondary check
What is the distribution?	histogram / KDE	box plot
How do categories compare?	bar/count plot	percentage table
How do two numerical variables relate?	scatter plot	correlation coefficient
Which variables move together?	heatmap	pair plot
How does target composition vary?	count/stacked bar	grouped percentage
Do groups have different score distributions?	horizontal bar plot	summary statistics
Need interactive exploration?	Plotly	hover/filter views

11. Python Tools and Libraries

Tool	Role in project
Pandas	data loading, cleaning, grouping, summaries
NumPy	numerical calculations and array operations
Matplotlib	custom static plots
Seaborn	statistical plots and heatmaps
Plotly	interactive charts
Jupyter Notebook	reproducible EDA workflow and narrative
Scikit-learn	later preprocessing, feature engineering and evaluation
Git/GitHub	version control and documentation

12. Detailed Feature-Engineering Readiness

Week 2 is EDA-focused, but findings will directly prepare later feature engineering. Every derived feature must use only information available at the prediction time.

Candidate feature	Example calculation	Why investigate?
Academic average	$(70 + 80) / 2 = 75$	summarizes multiple scores
Assessment gap	$78 - 65 = 13$	captures improvement/decline
Study-effort index	study hours ÷ absences (safe handling)	combines effort and attendance context
Attendance band	0–59, 60–74, 75–89, 90–100	tests nonlinear effects
Study-hours band	0–2, 3–5, 6–8, 9+	tests threshold effects
Interaction feature	attendance × study hours	captures combined behavior
One-hot encoding	support yes/no → 0/1	makes categories model-ready

Leakage prevention: future exam results, post-outcome interventions, or information unavailable at prediction time must not become predictors. The final feature list will be reviewed against the defined prediction point.

13. Model Evaluation Readiness

The planned later workflow is: clean → split data → fit preprocessing on training data only → train models → cross-validate → evaluate on an untouched test set.

Illustrative split: with 1,000 records, an 80/20 split produces 800 training rows and 200 test rows. If the illustrative target distribution is 70% Not At Risk and 30% At Risk, stratification would aim for about 560/240 in training and 140/60 in testing.

Example evaluation: if a model correctly predicts 180 of 200 test records, accuracy = $180/200 = 90\%$. Precision, recall, F1-score, and a confusion matrix will also be reported because identifying at-risk students may be more important than overall accuracy.

Metric	What it measures	Use in project
Accuracy	overall proportion correct	general performance
Precision	correct positive predictions / all positive predictions	reliability of at-risk alerts
Recall	correct positive predictions / all actual positives	how many at-risk students are detected
F1-score	balance of precision and recall	balanced metric
Confusion matrix	TP, TN, FP, FN counts	understand error types
5-fold CV	performance across five training folds	stability during model comparison

14. Reporting and Documentation Plan

Each EDA finding will use the structure **Question** → **Method** → **Result** → **Interpretation** → **Action**.

Example entry: Question: Is attendance associated with risk? Method: compare attendance distributions by target using box plots and grouped summaries. Result: calculate the actual difference from the dataset. Interpretation: determine whether the pattern is meaningful and whether outliers distort it. Action: decide whether attendance needs transformation, binning, or an interaction feature.

The final notebook/report will include dimensions, data dictionary, missingness table, duplicate count, outlier decisions, key plots, correlations, target distribution, important relationships, feature-engineering candidates, and limitations.

15. Expected Findings

- Identify the most frequent missing-value fields and quantify their rates.
- Determine whether numerical variables are skewed or contain extreme values.
- Measure target-class balance before selecting evaluation metrics.
- Identify candidate predictors with useful relationships to performance.
- Find redundant or highly correlated features that need review.
- Find feature-engineering candidates without introducing leakage.
- Create a reproducible visual record supporting later model decisions.

16. Challenges and Mitigation

Challenge	Potential impact	Mitigation
Missing values	biased summaries/model inputs	quantify first; choose justified imputation
Outliers	distorted mean/correlation	IQR + box plot + domain validation
Class imbalance	misleading accuracy	stratified split + precision/recall/F1
High correlation	redundant predictors	heatmap + feature review
Data leakage	overly optimistic model	define prediction point; fit transforms on train only
Too many plots	unclear report	tie each plot to a specific EDA question
Small categories	unstable conclusions	report counts and group carefully

17. 32-Hour Week 2 Work Plan

Activity	Hours	Output
EDA planning and data dictionary preparation	3	EDA checklist + variable map
Dataset inspection and quality profiling	5	shape/types/missing/duplicate report
Univariate analysis	4	distribution plots + summaries
Bivariate analysis	5	relationship plots + target comparisons
Multivariate analysis	4	heatmap/interactions/grouped analysis
Outlier and missing-data strategy	3	documented treatment rules
Visualization and notebook formatting	3	organized charts + captions
Findings, feature-engineering readiness and documentation	5	final EDA report/notebook
Total	32	Completed Week 2 framework

18. Feedback-Alignment Checklist

Feedback item	How this revised Week 2 addresses it
Specific examples	Includes 1,000-record examples, 40/1,000 = 4% missingness, 70:30 class balance, 80/20 split, and calculation exam
Concrete numbers	Includes 4% missingness, 700/300 class example, 800/200 split, 560/240 and 140/60 stratified example, and 90% ill
Diagrams included in text	EDA workflow and visualization strategy diagrams are embedded directly in the PDF with captions.
Detailed feature engineering	Covers averages, gaps, bins, interactions, encoding, and leakage prevention.
Detailed model evaluation	Covers evaluation sequence, metrics, confusion matrix concepts, stratification, cross-validation, and untouched test s
Avoid unsupported claims	All numerical values are clearly labeled illustrative/planning assumptions; actual results will only be reported after dat

19. Conclusion

This revised Week 2 framework turns EDA into a measurable and documented process rather than a list of generic plots. It defines what will be checked, which visualization answers each question, how numerical findings will be interpreted, and how EDA outputs will feed into feature engineering and model evaluation.

Important note: All numbers in this document are illustrative planning examples. They are not claimed experimental findings from the final dataset.